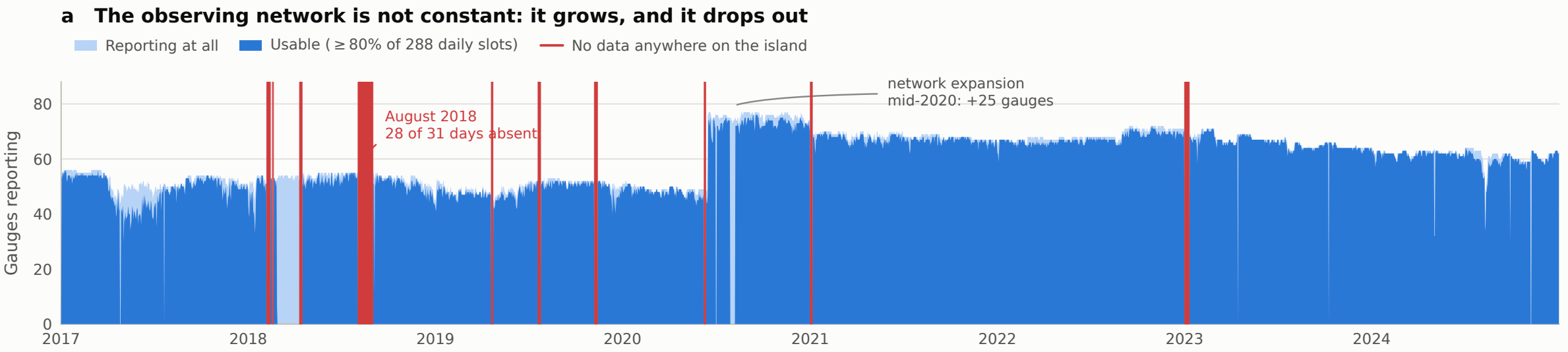
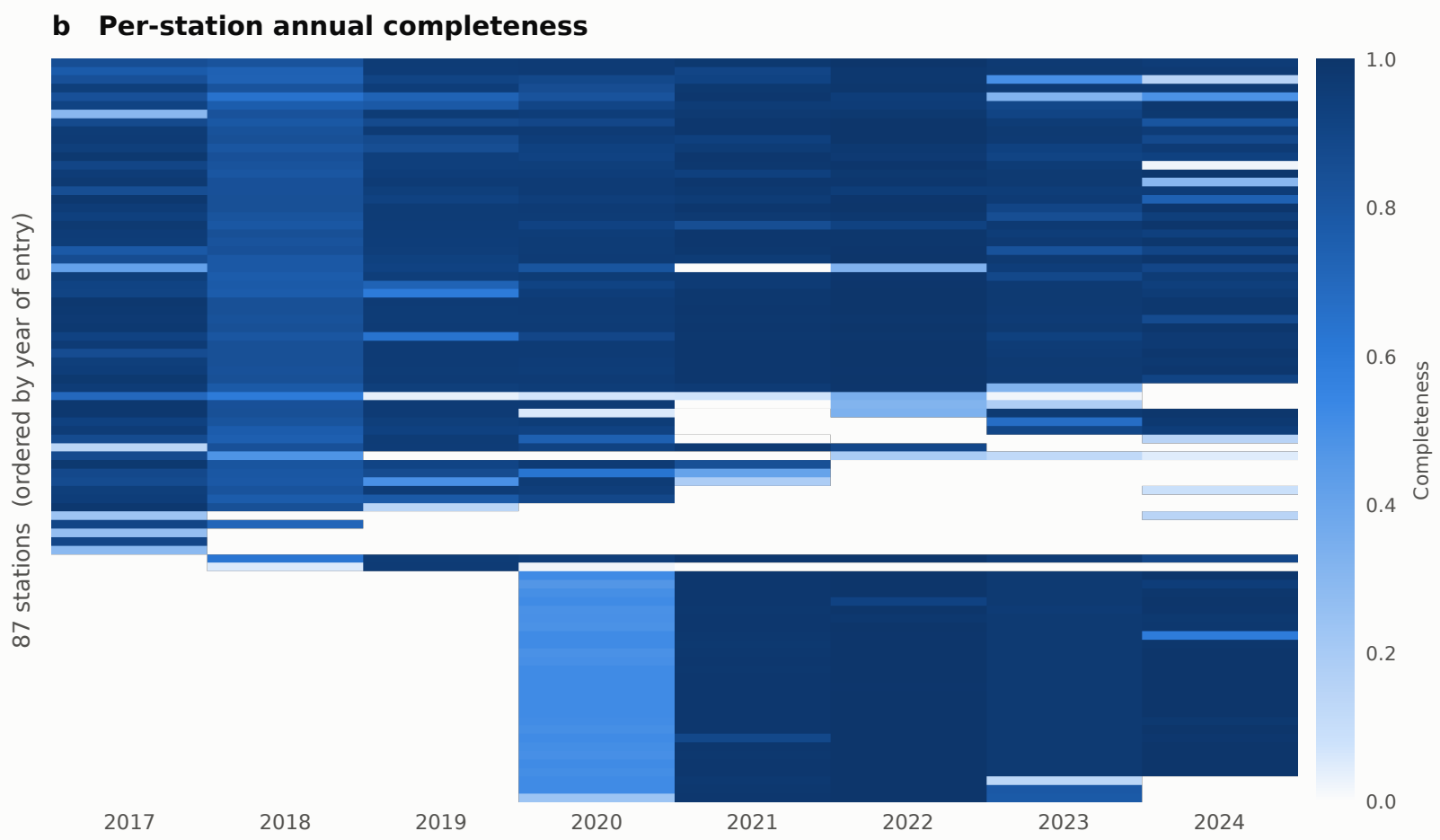
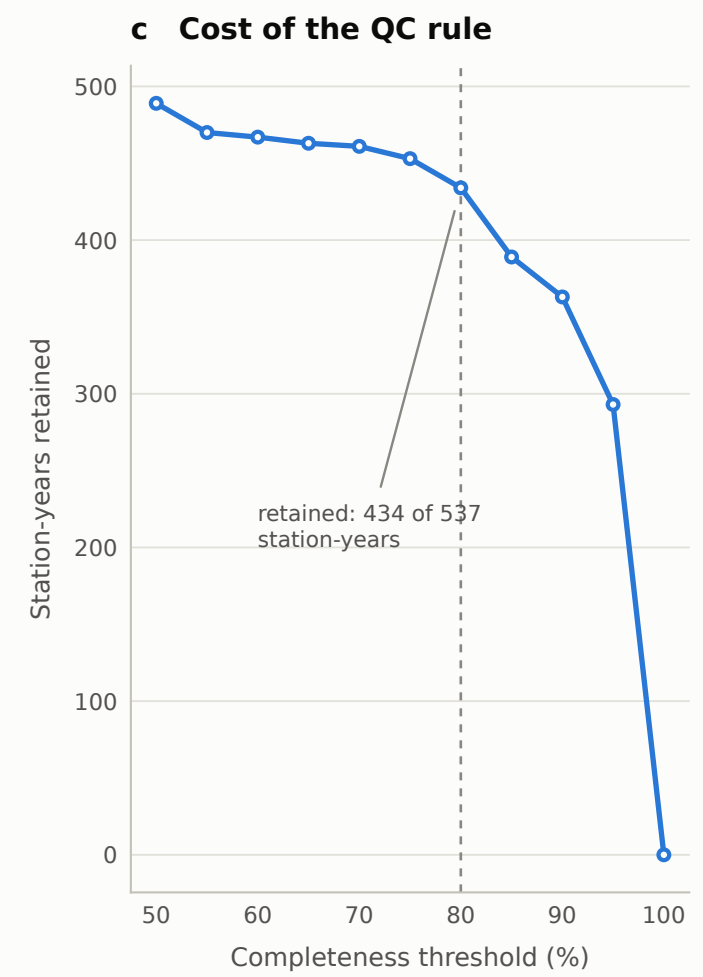
# A changing gauge network, not a changing climate, is the first thing the raw data shows

Before any weather signal can be read, the record has to be audited. The network triples in places and vanishes in others, so island-wide figures in this report are areal means over gauges that pass a completeness rule, never raw averages over rows.



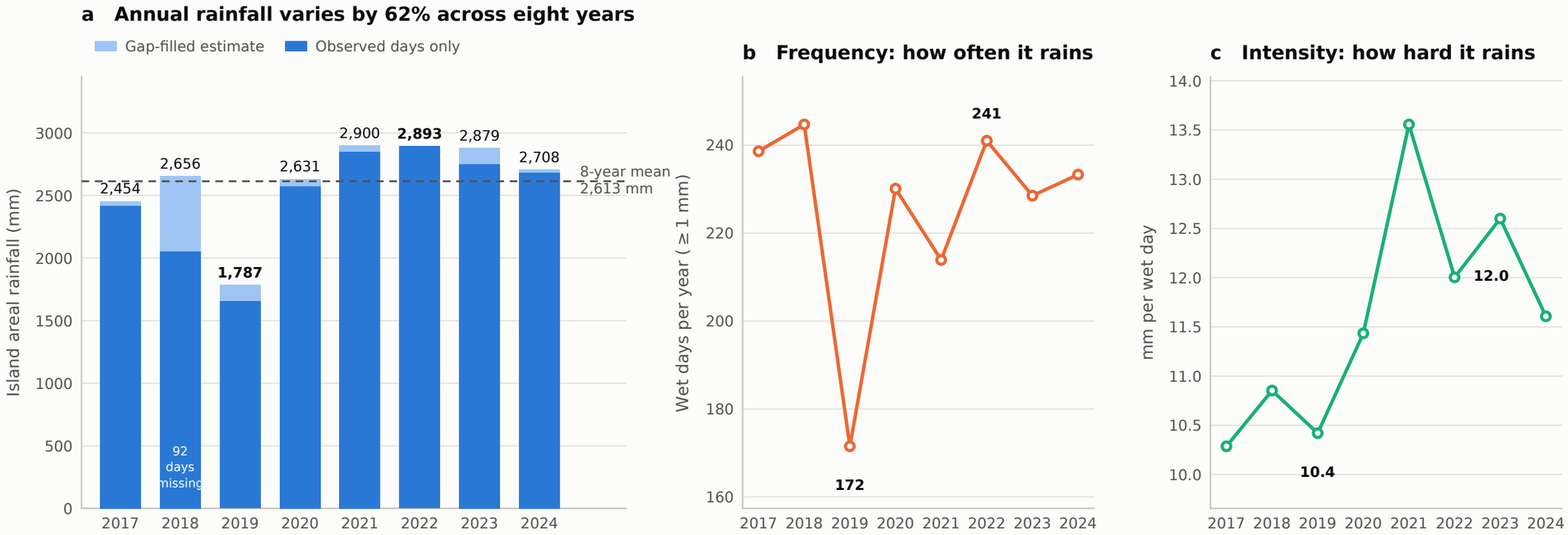




Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.
QC rule: a station-day needs ≥231 of 288 five-minute slots (≥80%); an island-day needs ≥20 such stations.

# 2019 was dry because it rained less often, not because the rain got lighter

Splitting the annual total into frequency × intensity separates the two ways a year can be dry. The 2019 deficit (1,787 mm; it coincides with a positive Indian Ocean Dipole in external climate records) is a collapse in wet-day count, 172 days against a 233-day average for the other years, while the rain that did fall was no lighter than 2017's. Panel c carries a separate signal:
per-wet-day intensity rises sharply after 2019 and remains generally above the 2017–2019 level (12.2 vs 10.5 mm), although it varies from year to year.
Panels b and c are drawn on their own axes rather than twinned onto panel a.

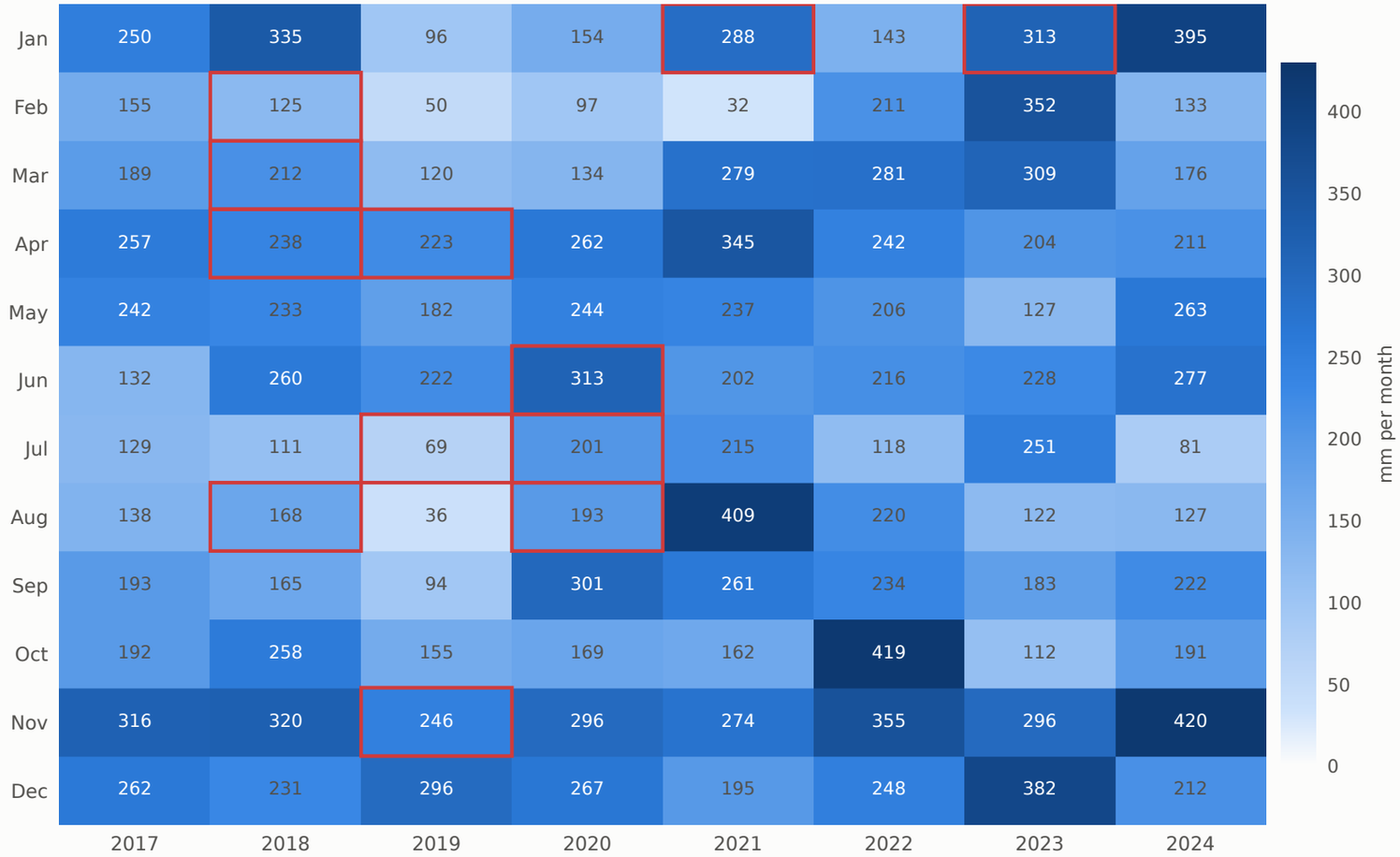



Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.
Wet-day counts are rescaled to a 365-day year so that years with missing days stay comparable.

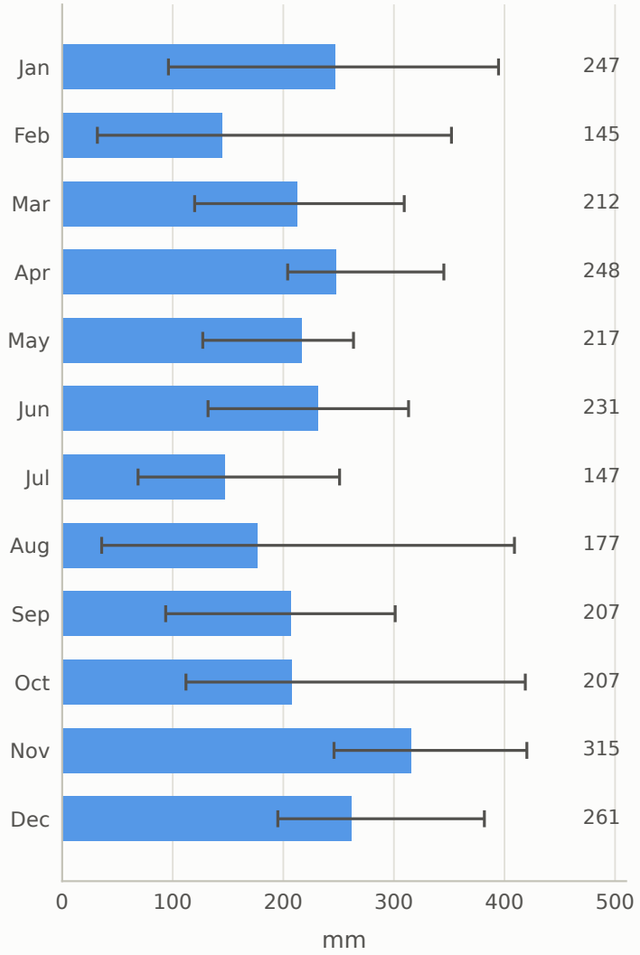
# A double wet season, and a year-to-year spread that swamps it

November–January (north-east monsoon season) and April–May (inter-monsoon) are the wet peaks; February and July are the dry troughs. But the same calendar month can differ by a factor of ten between years: February spans 32 mm (2021) to 352 mm (2023). Any monthly climatology for Singapore is therefore a weak predictor of any individual month.

## a Monthly island rainfall, 2017–2024 (mm)



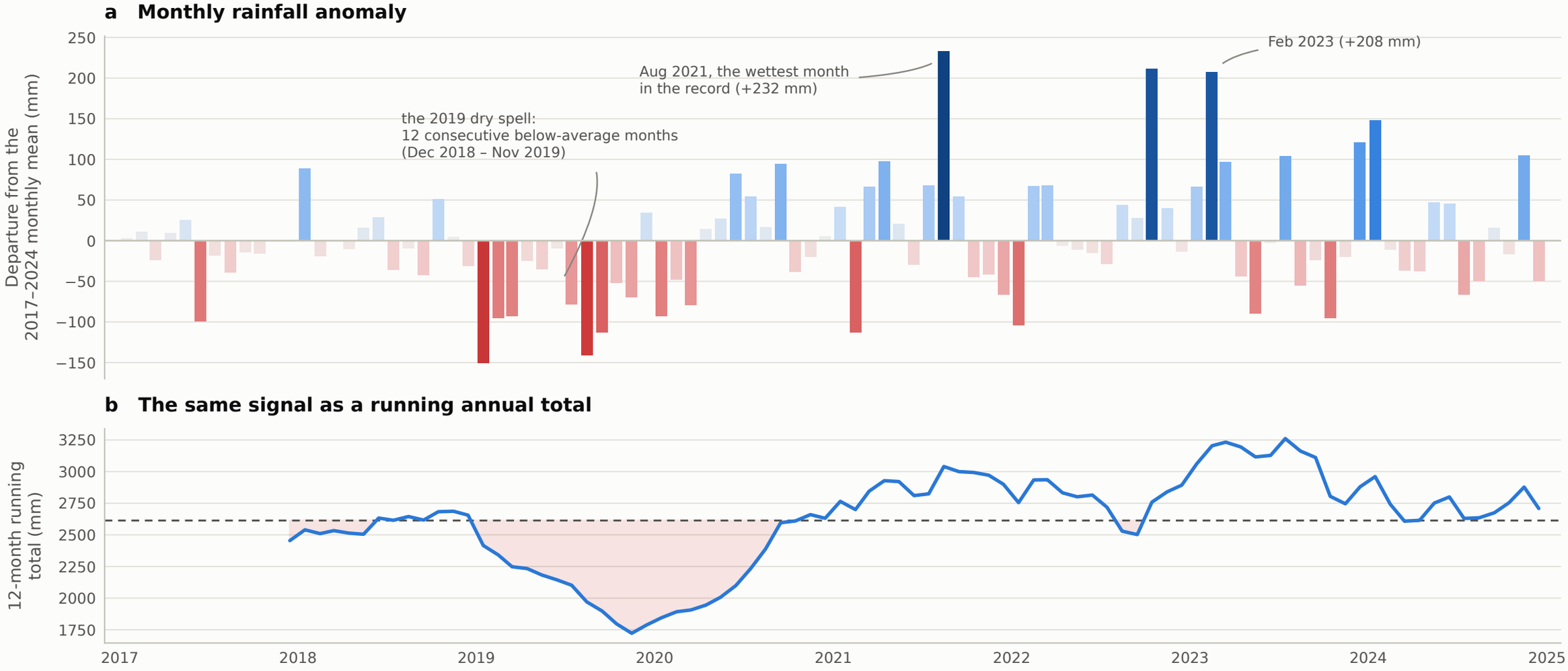

## b Climatology (bars) and the 2017–2024 range (whiskers)




Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.
Red outline: month whose source record is <90% complete, filled from the (month, day) climatology of the other years.

# Wet and dry spells in Singapore run for seasons, not for months

Anomalies are strongly autocorrelated: the 2019 dry spell runs 12 consecutive below-average months (the sustained 2021–2022 surplus coincides with La Niña conditions in external climate records). The run length is sensitive to one marginal month: on a fixed 21-gauge network 11 of those 12 months are still below average. The persistence is why the year-to-year spread in Figure 3 is so large, and a reason a short-horizon forecast (Task 2) should be given recent-rainfall state, not just a calendar position.

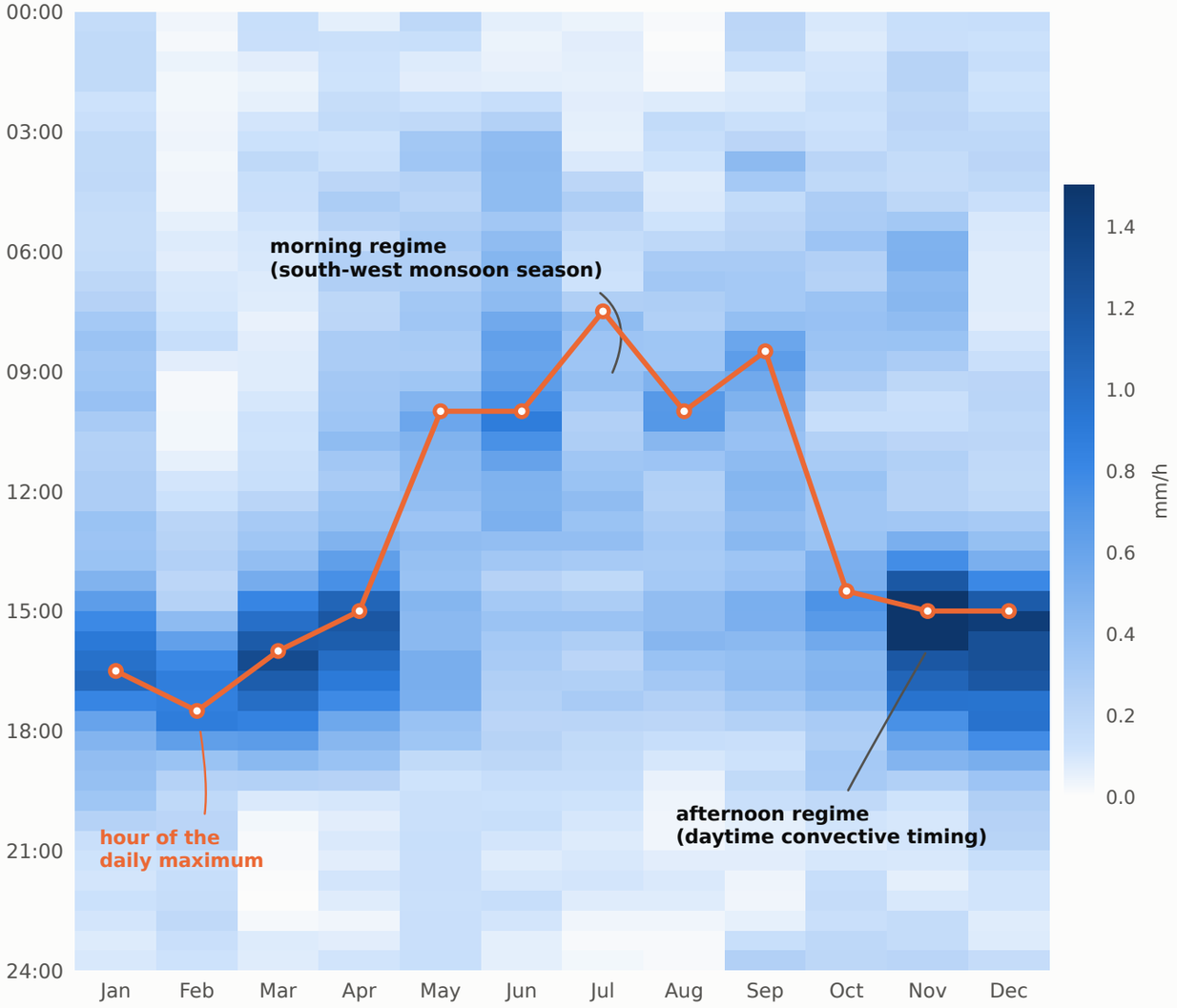



Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.
Anomalies are computed against each calendar month's own 2017–2024 mean, from the gap-filled island daily series. Panel b needs 12 months of data (starts Dec 2017).

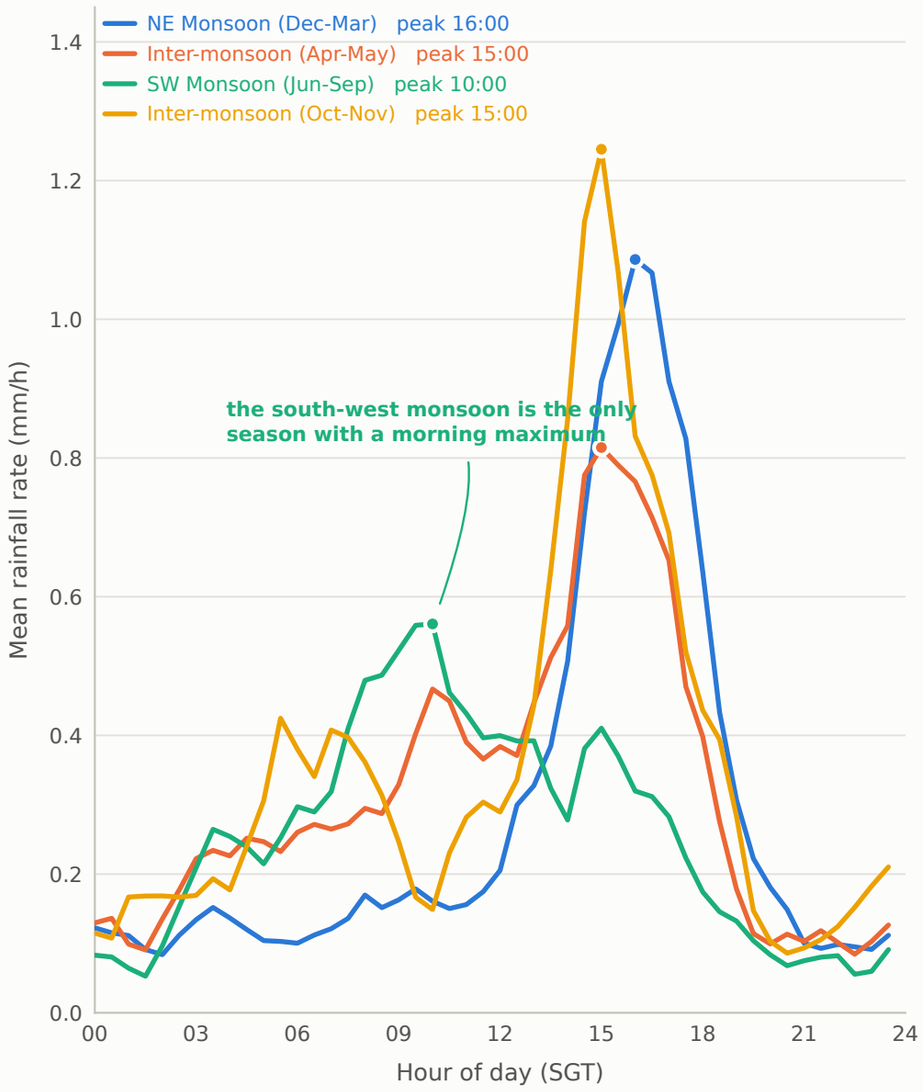
# Singapore has two rainfall clocks, and the monsoon decides which one runs

From June to September (south-west monsoon season) the daily rainfall maximum moves into the morning. The literature links this timing to Sumatra squall lines that develop over the Malacca Strait and reach Singapore around dawn (external source to be cited; not tested with this dataset). In every other season the maximum is a sharp mid-afternoon peak near 15:00–16:00, the timing expected of daytime convection. The peak hour swings by roughly six hours between the two regimes.

## a Mean rainfall rate by time of day and month (mm/h)




## b The clock changes with the monsoon

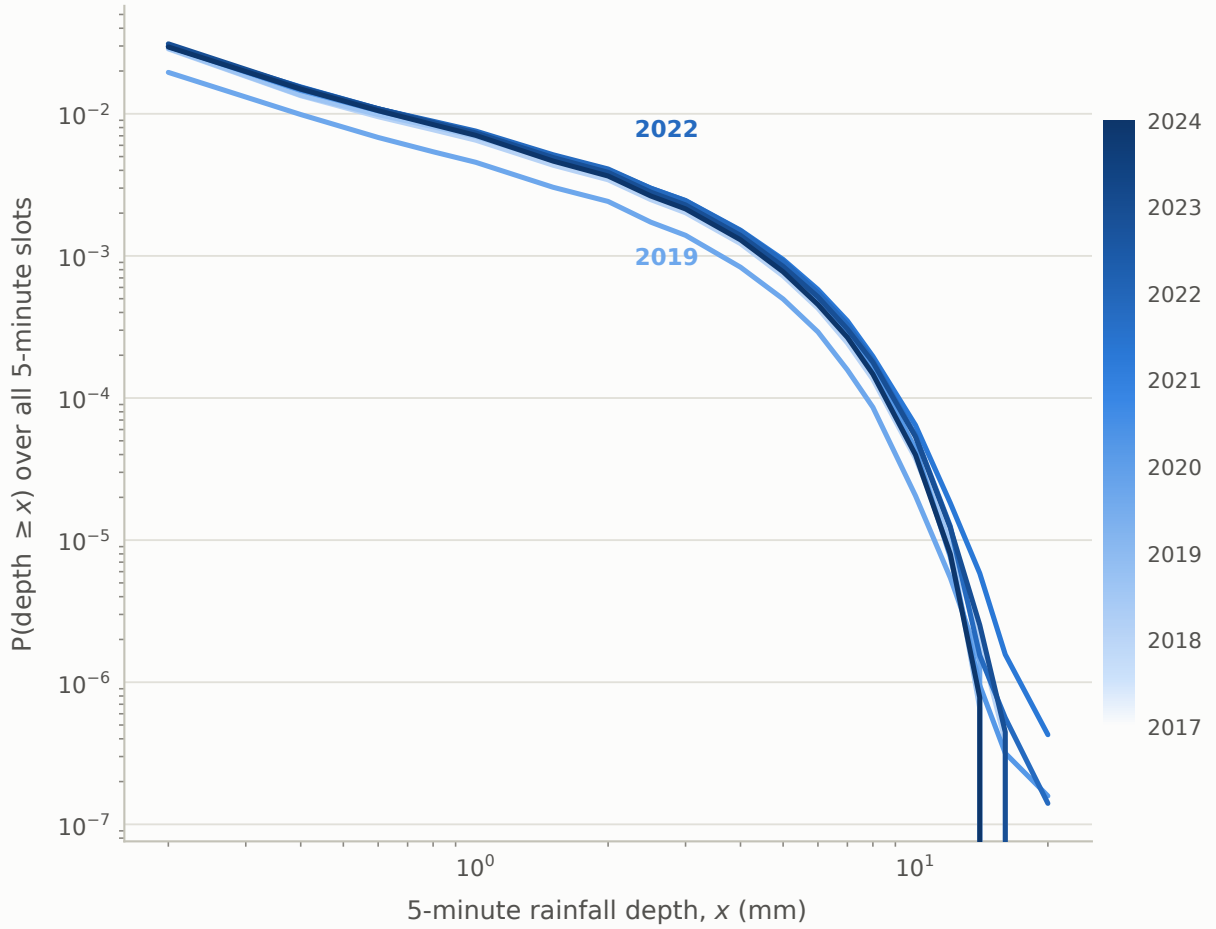



Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.
Rates are means over all stations and all 5-minute slots in each half-hour bin (dry slots included), so panel b is a true expected rate.

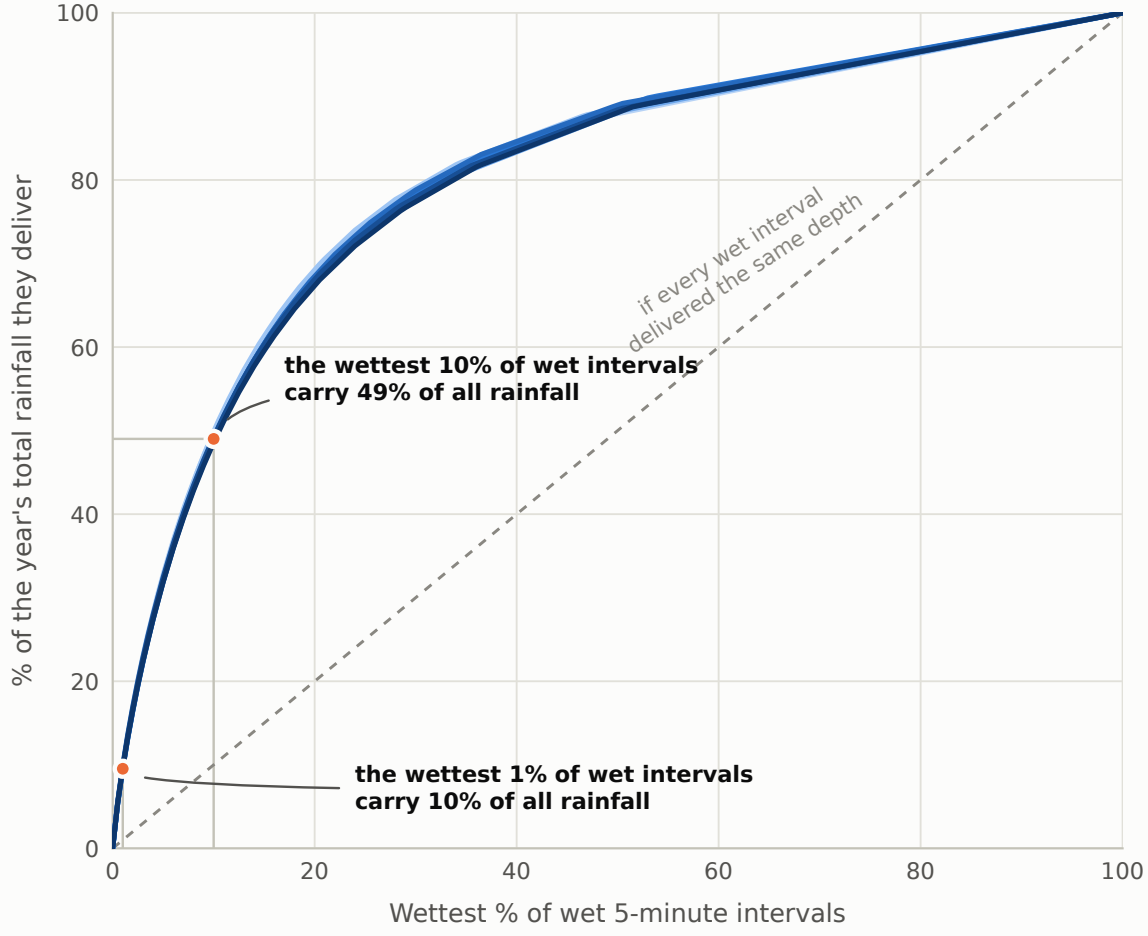
# Almost half the rain that falls on Singapore arrives in 10% of wet 5-minute intervals

The concentration curve is near-identical in every year, including the drought: a dry year is not a year of gentler rain, it is a year with fewer rain events. The heavy tail in panel a is why a rainfall model should be framed as an exceedance probability rather than a conditional-mean regression, which is exactly how Task 2 is set up.

## a Intensity is heavy-tailed, and the whole curve moves with the year




## b Rainfall is extremely concentrated in time

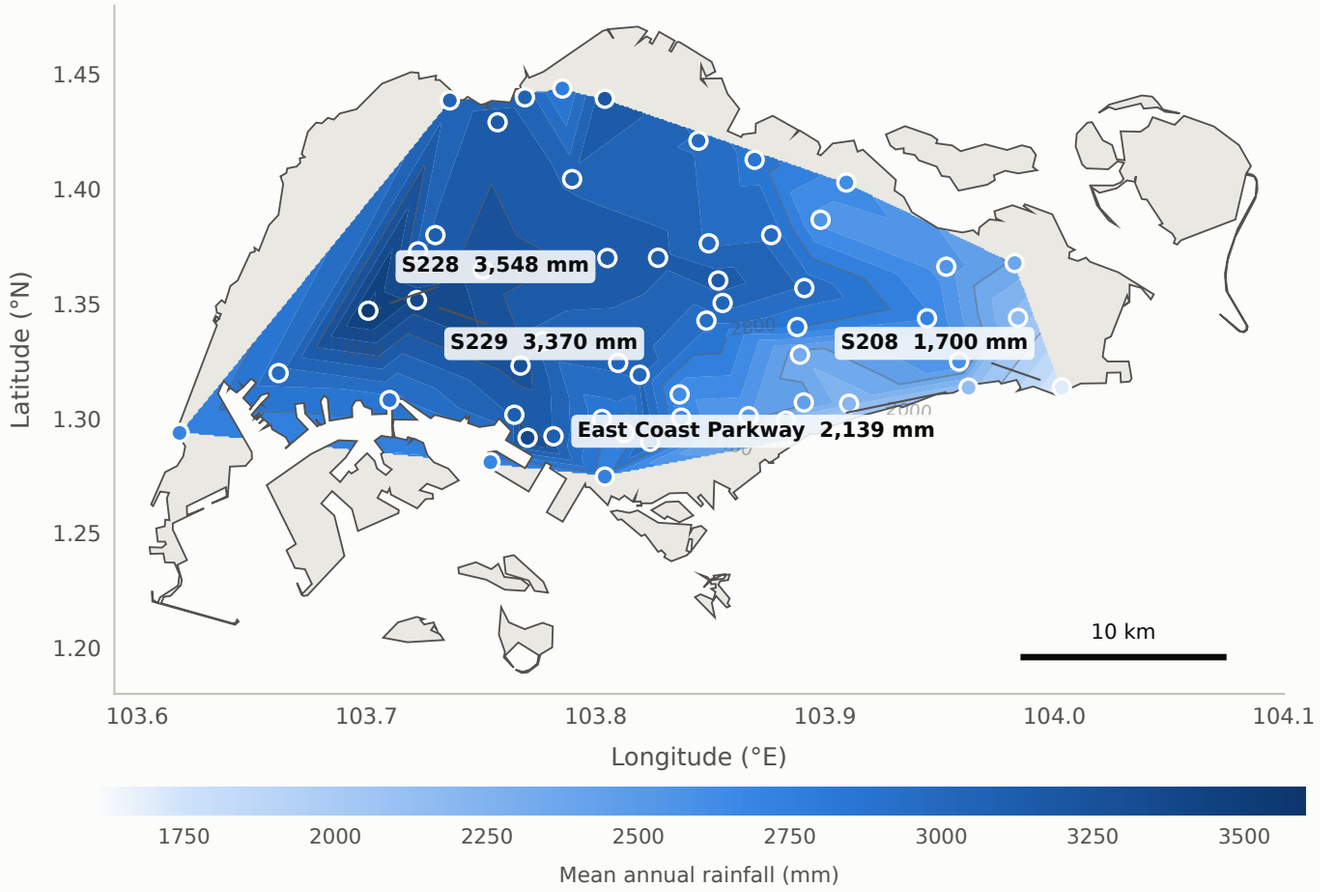



Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.
"Wet" = a 5-minute slot recording ≥0.2 mm (one tipping-bucket resolution step).

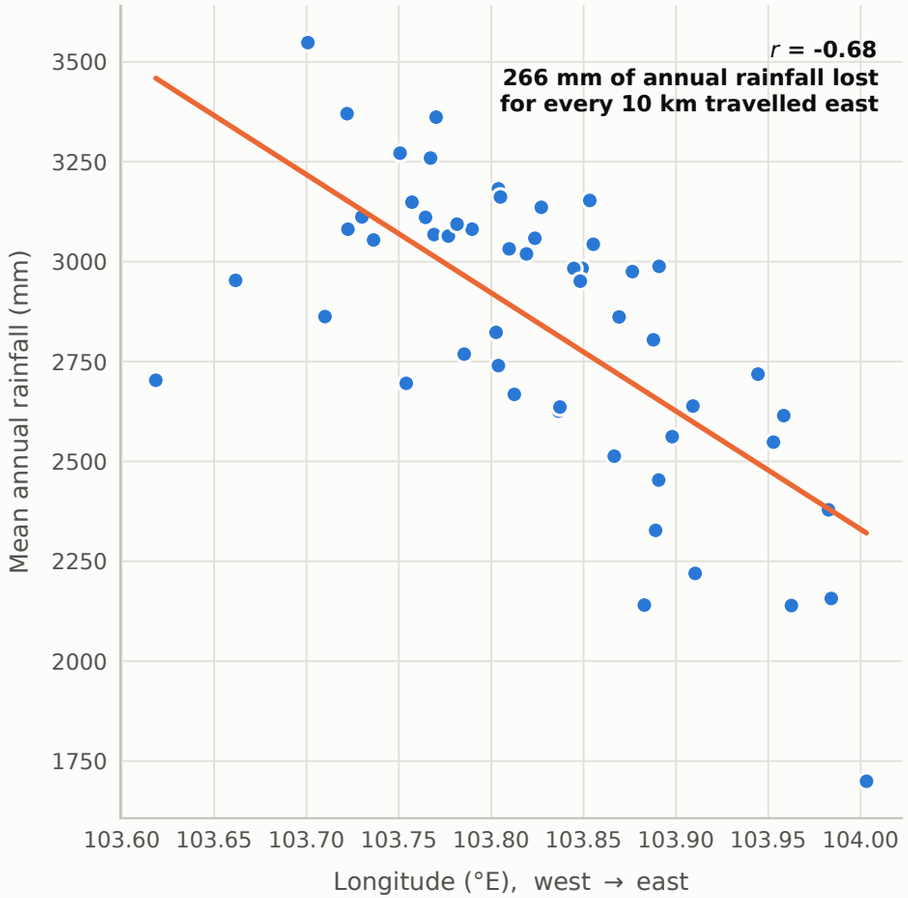
# Rainfall more than doubles across an island 50 km wide

Over the common 2021–2024 period the wettest gauge records 3,548 mm a year and the driest 1,700 mm, a factor of 2.1. The wet core sits over the western and central parts of the island and the east and offshore islands are markedly drier; sea-breeze convergence and terrain are commonly proposed explanations in the literature but are not tested here. Station identity therefore carries real predictive information, and Task 2 keeps location as a feature.

**a Mean annual rainfall, 2021–2024 (53 gauges with complete records)**



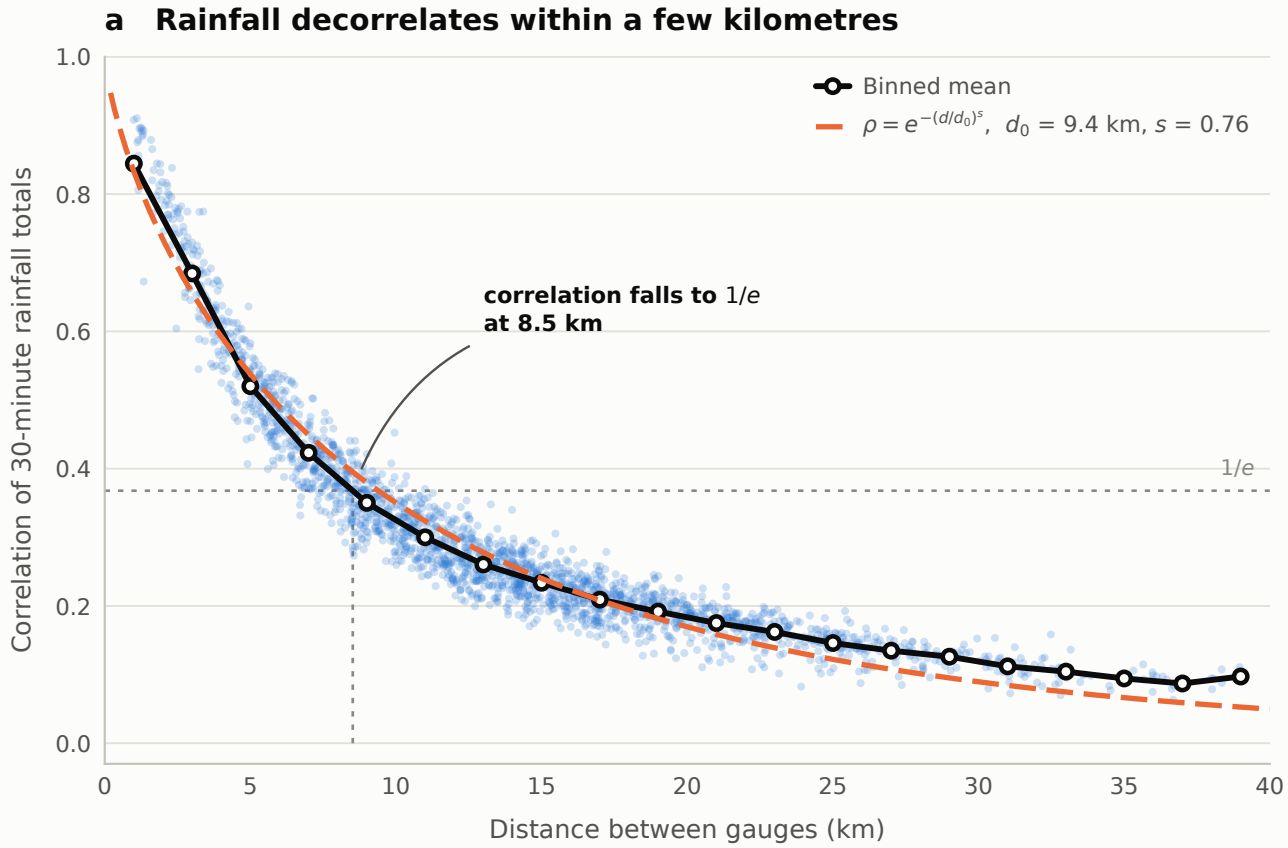

**b A steep west–east gradient**

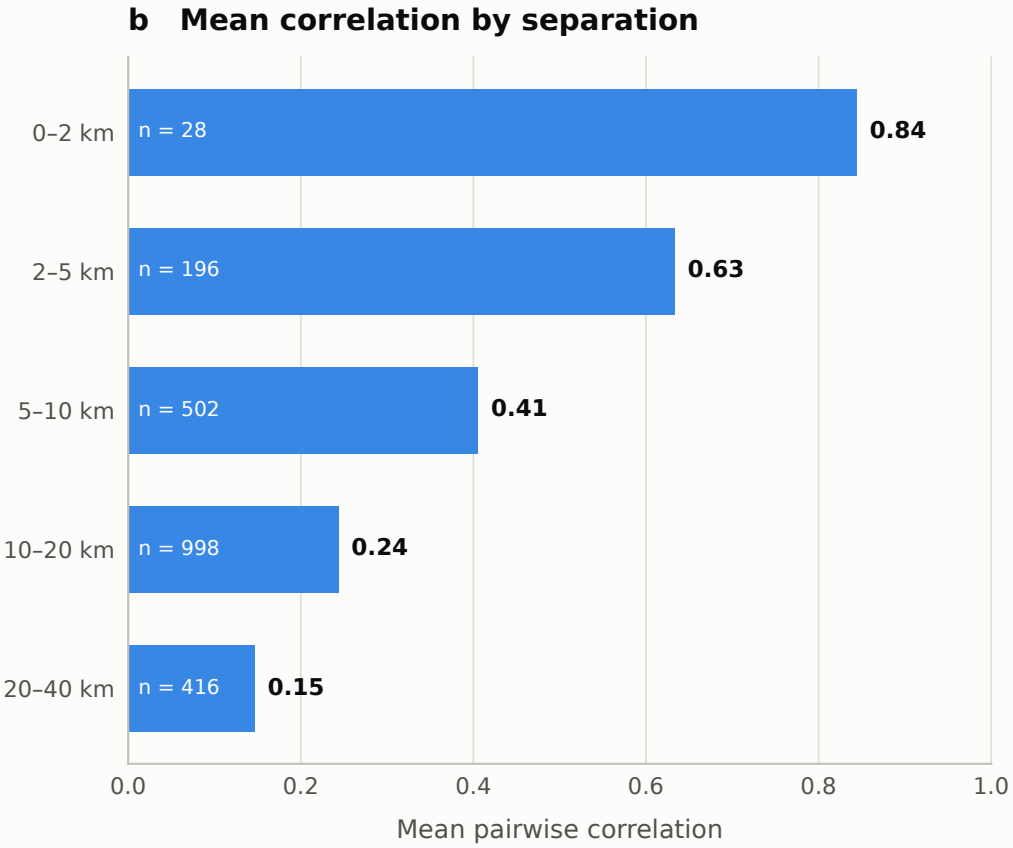



Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.
A fixed four-year window is used so that gauges commissioned at different times are never compared over different periods. The field in panel a is a linear interpolation between gauges, shown on land only and not extrapolated beyond the gauges' convex hull (grey land: no estimate). Coastline: URA Master Plan 2014 via geoBoundaries (ODbL 1.0).

# Singapore rainfall loses coherence over about 9 km, a fraction of the island's width

Gauges less than 2 km apart agree closely on their 30-minute totals ($\rho$ = 0.84); beyond 20 km they are nearly independent ($\rho$ = 0.15).
That is consistent with convective, cell-scale rainfall rather than broad frontal systems, and it carries a concrete modelling consequence:
a single island-wide forecast cannot be right everywhere at once, so Task 2 predicts per station and draws on neighbouring gauges only at short range.



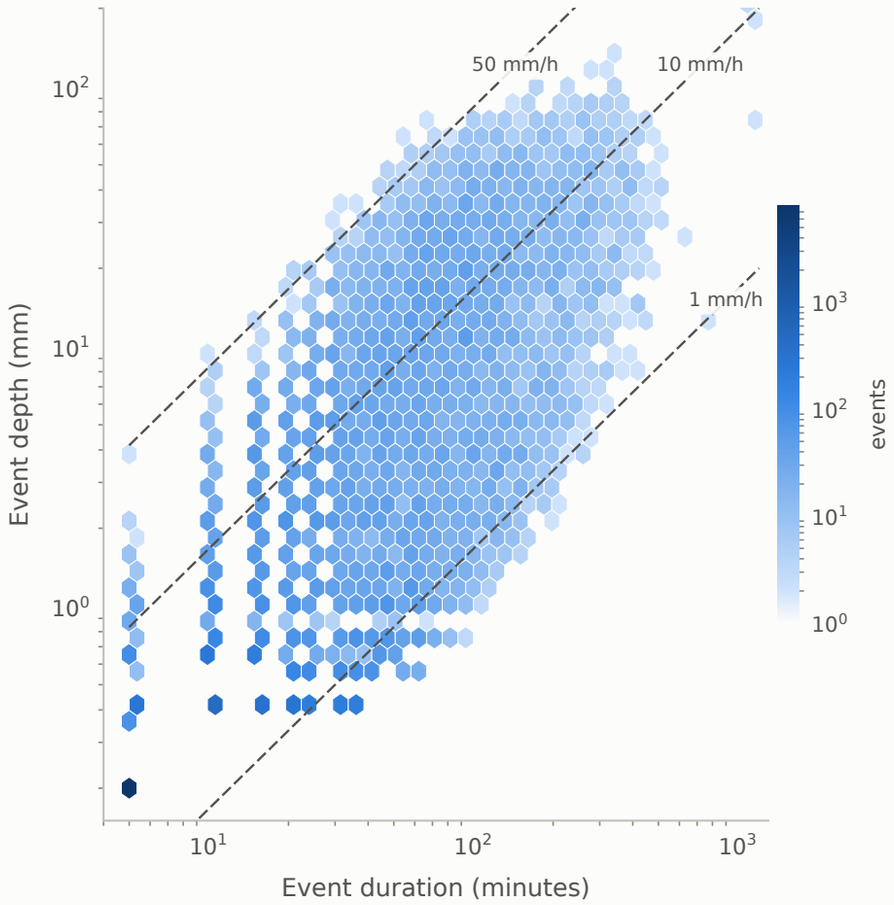




Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.
2,145 station pairs, Pearson correlation of concurrent 30-minute totals over 2017–2024 (66 gauges with ≥4 complete years; pairwise-complete observations, ≥5,000 overlapping half-hours per pair). The 8.5 km figure is read off the binned means and assumes no functional form.

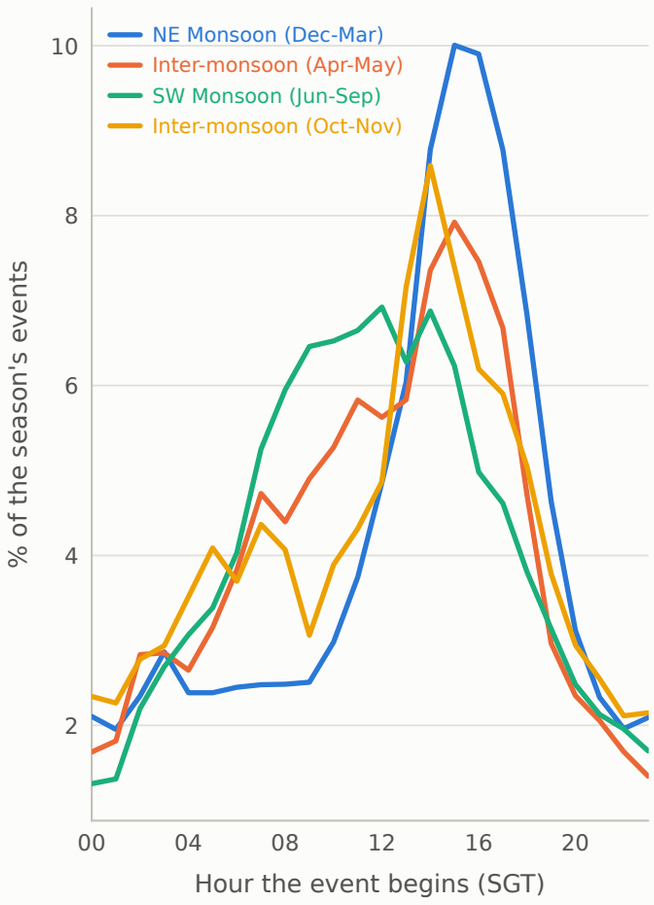
# The typical rainfall event is short, light, and begins when the season says it should

Splitting each gauge's record into events separated by ≥30 minutes of dry weather gives a median duration of 20 minutes and a median depth near 1 mm, with a long tail of hour-scale storms along the 10–50 mm/h lines in panel a. Panel b recovers the same two regimes as Figure 5 from a completely different construction, and panel c shows the 2019 drought as a deficit of events, not of event size.

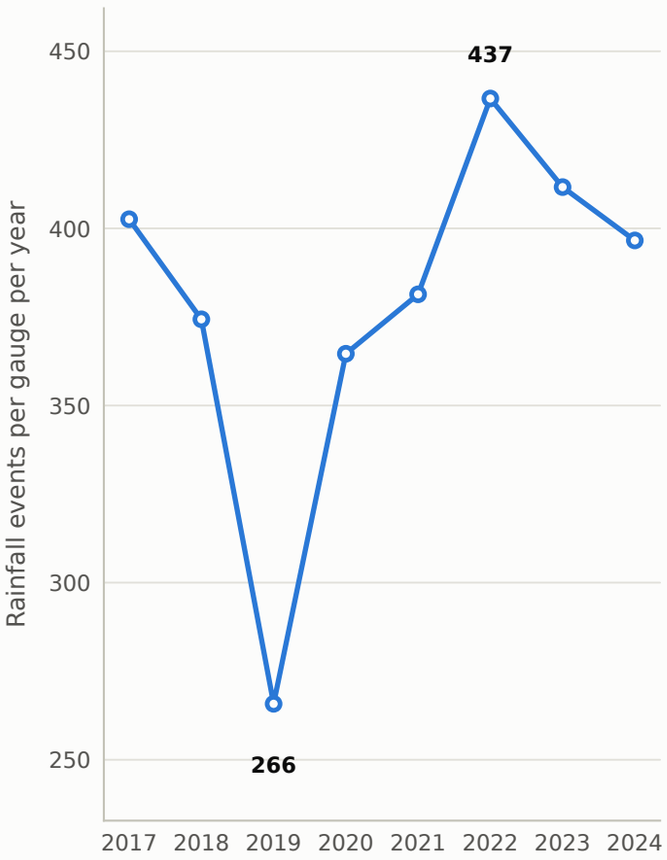
## a Event duration vs depth




## b When events start



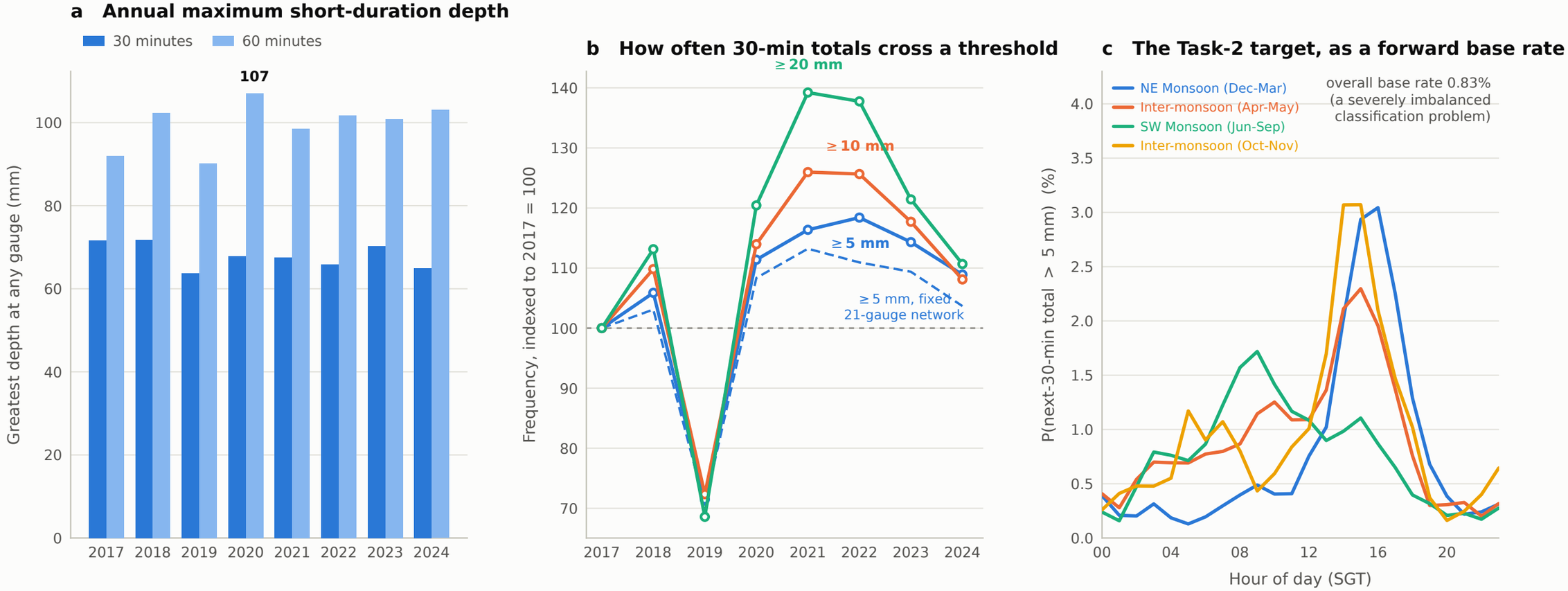

## c Event frequency drives the dry years




Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.
An event is a maximal run of wet 5-minute slots at one gauge with no dry gap exceeding 30 minutes. Panel a plots a 12% random sample for legibility.

# Heavy half-hours were about 23% more frequent in 2020–2024 than in 2017–2019, and they keep the same daily rhythm

Panel b indexes the frequency of 5, 10 and 20 mm trailing half-hour totals to 2017. All three rise into the wet 2021–2022 period and stay elevated. On a fixed set of 21 gauges (dashed) the ≥5 mm rise is 20%, so the 2020 network expansion explains little of it; the 2019 drought in the baseline explains more, and over eight years it is not a significant trend (Mann–Kendall p = 0.28). The annual maxima in panel a show no trend either (p = 0.40).
Panel c is the quantity Task 2 must estimate: the probability that the NEXT 30 minutes (strictly) exceed 5 mm, by hour of the prediction time and season.




Source: data.gov.sg, Historical Rainfall across Singapore, 2017–2024 (48.5 M five-minute gauge readings, 91 stations). Processed with Apache Spark.
Panels a, b: trailing 30- and 60-minute accumulations per gauge (history). Panel c: forward window t to t+30 min, six complete 5-minute slots required, strict ">".